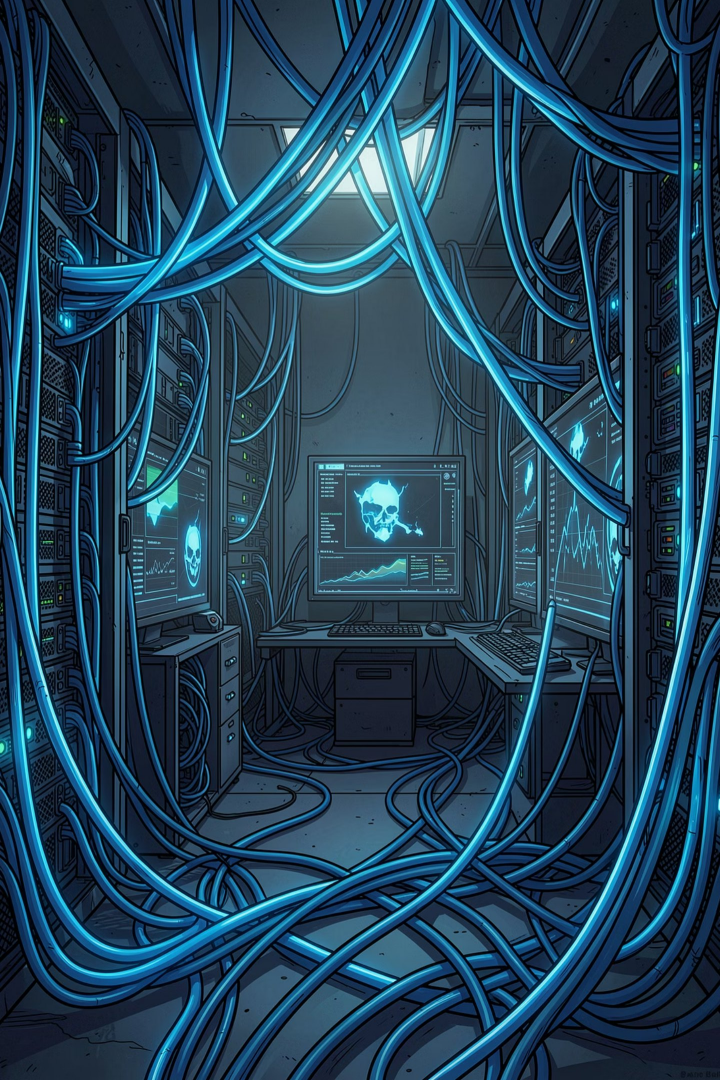




Hybrid Multi- Interface IDS/IPS Platform Using Suricata

Behavioral analysis and real-time threat detection system

NETWORKSECURITY

INTRUSION DETECTION

AUTOMATED RESPONSE

Project Introduction

An integrated platform for monitoring network traffic across multiple interfaces, combining real-time threat detection, behavioral analysis, and automated response.



Network Monitoring

Simultaneous monitoring of LAN and WAN traffic



Attack Detection

Behavioral analysis and detection of malicious activity as it happens



Instant Alerts

Real-time notifications when any threat is detected



Automated Response

Automatically blocking attackers via iptables

Technologies used: Suricata · Python · Flask · iptables · USB WiFi Adapter



The Problem: Limitations of Traditional IDS Systems

Traditional intrusion detection systems suffer from structural gaps that make them insufficient to address modern threats.

Monitors only one interface

Unable to track data traffic across multiple network interfaces at the same time

Static detection with no behavioral analysis

Relies entirely on fixed signatures without analyzing suspicious behavior patterns

Slow manual response

Requires human intervention to take action, which prolongs the exposure window to attacks

Lack of wireless network monitoring

No support for monitoring wireless interfaces or detecting intrusion over WiFi

The original project and its limitations

The initial system relied on monitoring the internal LAN network only, with basic Suricata rules and fixed alerts without any operational intelligence.

Key gaps and limitations



No monitoring of WAN traffic or external internet traffic



No automatic blocking of attackers or malicious addresses



No detection of wireless intrusion via WiFi



No system for recording or classifying threat scores

LAN only
monitoring

Fixed alerts

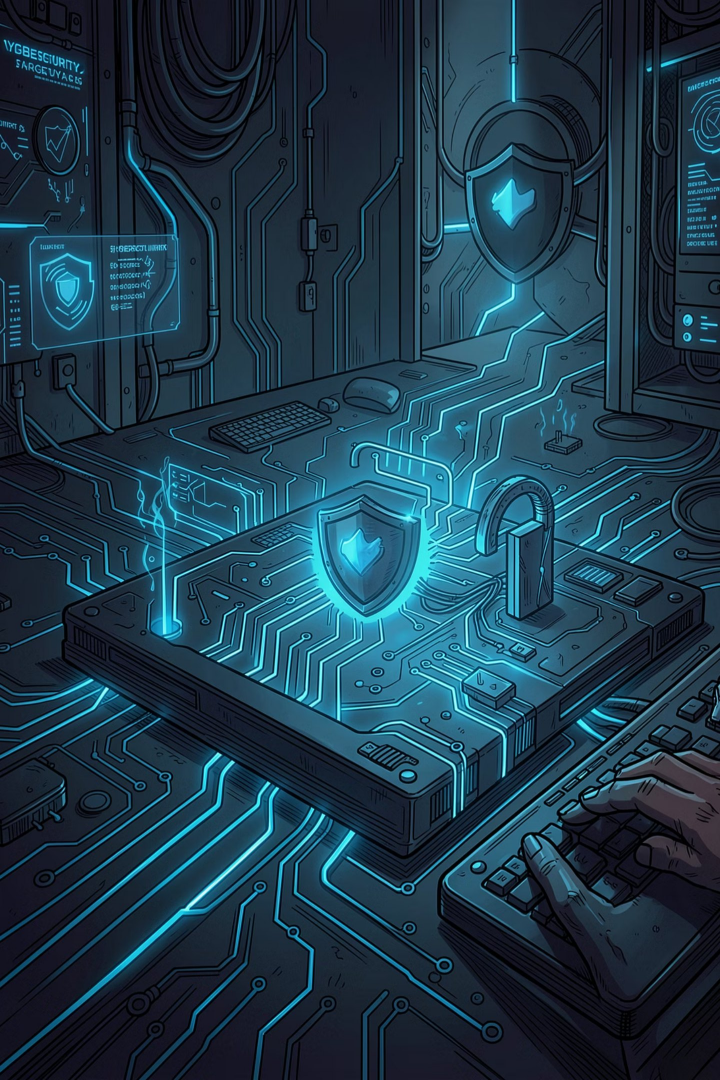
Basic Suricata
rules

Manual
monitoring

Project Evolution: From the Original System to the Hybrid Platform

A significant leap in performance and capabilities through expanding the monitoring scope and adding AI-driven intelligence.

Dimension	Original System	Enhanced System
Monitoring scope	LAN only	LAN + WAN + wireless
Detection method	Static signature-based detection	Advanced behavioral analysis
Alerts	Basic alerts	Threat scoring
Response	Fully manual	Instant automated blocking
Interfaces	Single wired interface	Multiple wired and wireless interfaces



Project Objectives

01

Comprehensive Monitoring

Monitor LAN and WAN traffic simultaneously and continuously

02

Attack Detection and Behavior Analysis

Identify malicious activity and analyze attacker patterns

03

Alerts and Automated Blocking

Instant notifications and automatic blocking of dangerous sources

04

Wireless Detection Support

Monitor WiFi networks and detect wireless intrusion



Why Suricata?

Suricata is an open-source IDS/IPS engine distinguished by high performance and advanced capabilities, making it the ideal choice for this platform.

High-performance multi-threaded engine

Processes network packets efficiently without affecting performance

Real-time packet inspection

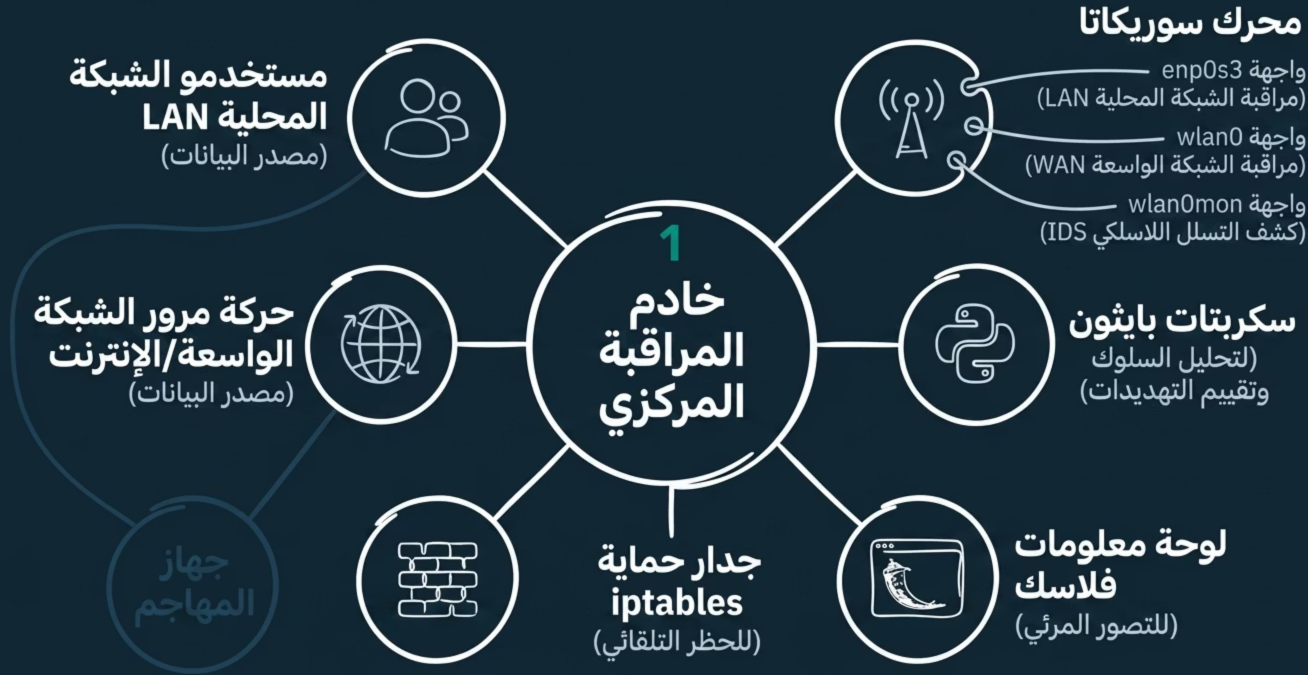
Deep Packet Inspection with support for multi-protocol analysis

Custom and flexible rules

Full support for writing and applying custom detection rules according to the environment

System Architecture

An integrated design that connects monitoring, analysis, and response components into a unified system.



Interfaces: `enp0s3` ← LAN monitoring, `wlan0` ← WAN monitoring, `wlan0mon` ← wireless IDS

Multi-Interface Monitoring



LAN Monitoring —
`enp0s3`

Monitor internal user traffic, detect horizontal attacks and lateral movement within the local network



WAN Monitoring —
`wlan0`

Analyze external internet traffic, monitor inbound and outbound connections, and detect outside threats



Wireless IDS —
`wlan0mon`

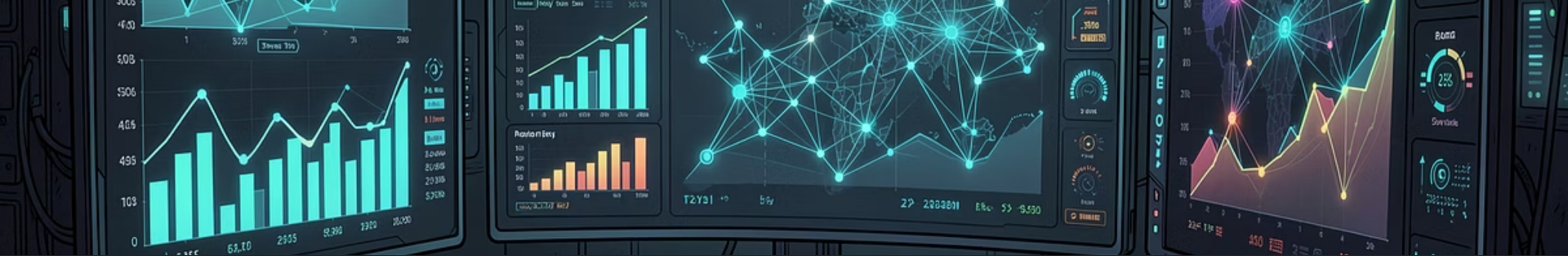
Monitor wireless networks in monitoring mode, detect rogue access points and WiFi attacks

Comparison: LAN Monitoring vs WAN Monitoring

Each interface serves a different but complementary security purpose to provide comprehensive coverage.

Comparison Criterion	LAN Monitoring	WAN Monitoring
Traffic Type	Internal network traffic	External internet traffic
Source	Local users	Internet traffic and remote servers
Threat Types	Lateral movement, internal leakage	External attacks, internet-based intrusion
Interface	enp0s3	wlan0
Monitoring Scope	Closed local network	Network boundary with the internet
Security Priority	Prevent internal threats	Repel external attacks

 Both interfaces operate simultaneously under Suricata supervision to ensure comprehensive 360° security coverage



Comprehensive Security Dashboard

Centralized interface for real-time visibility and proactive threat management.



Threat Statistics

Visual distribution of threat categories and attack vectors.



Live Alerts

Instant notifications for critical security events and anomalies.



Active Attackers

Identification of suspicious IPs with threat scores for quick triage.



Blocked IPs

Complete records of automatically blocked malicious sources with reasons.



Traffic Analysis

Interactive charts to visualize network patterns and detect unusual flows.

Threat Detection Timeline

Our platform responds to emerging threats with speed and precision, progressing through detection, analysis, and automated response.



Real Attack Scenario: Detection to Defense

Our platform swiftly processes threats from initial detection to automated blocking, ensuring rapid response and containment within seconds.



Suspicious Upload

Malicious file attempts to enter the network



THREAT Security



Suricata Detection

Signature-based rules flag the activity.



Behavioral Analysis

Python scripts analyze patterns for malicious intent.

THREAT AMY



Threat Score Increases

Risk calculated based on multiple indicators.



High Classification

System confirms critical security risk



Automated Blocking

iptables instantly isolates and blocks the source.

Real-Time Monitoring and Wireless Detection

Our platform ensures continuous vigilance through advanced real-time monitoring and specialized wireless intrusion detection capabilities.

Real-Time Monitoring



fast.log Logs Detailed logging of network events for rapid incident response and analysis.



ICMP and Uplink Detection Instant alerts on suspicious ICMP traffic and network uplink status changes.



Attack Logging Comprehensive records of detected attack attempts, including source, target, and type.

Wireless Intrusion Detection



USB WiFi Adapter (wlan0mon) Dedicated interface for passive wireless network monitoring in monitor mode.



Aircrack-ng Tools Integration Utilizes specialized tools for advanced wireless network analysis and attack identification.



Packet Capture Captures wireless packets to detect rogue access points, deauthentication attacks, and other threats.

Attack Classification

Detailed classification of detected threats allows for a more precise understanding of attack vectors and tailored response strategies within the IDS/IPS platform.



Suspicious Uploads

Detection of unusual file transfers, large volumes of data uploaded, or executables in unexpected locations, indicating data exfiltration or malware staging.



Flood Attacks

Identification of excessive network traffic aimed at overwhelming system resources, such as DDoS attempts, SYN floods, or UDP floods.



Wireless Threats

Recognition of attacks specifically targeting Wi-Fi networks, including rogue access points, deauthentication attacks, or brute-force attempts on wireless passwords.



Repeated Attempts

Tracking of multiple failed login attempts, port scans, or connection attempts from suspicious IPs, often precursors to brute-force or reconnaissance attacks.



Malicious Files

Detection of known malware signatures, exploit kits, ransomware, or other harmful payloads being transferred or executed within the network.



Stealth Indicators

Identification of evasive techniques such as encrypted traffic anomalies, covert channels, or unusual port usage designed to bypass traditional security measures.

Automated Threat Scoring

Our system assigns a dynamic threat score to detected events, enabling rapid prioritization and response based on severity.

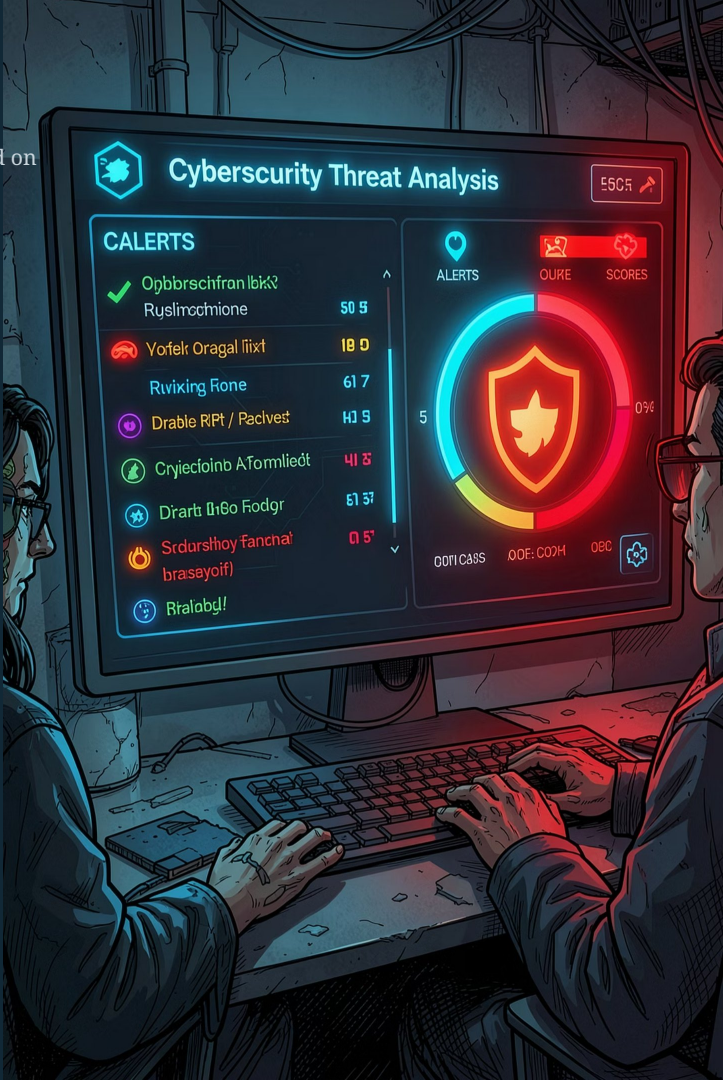
Event Scoring Rules

Event Type	Score
Normal upload	0
Repeated upload	+2
Suspicious file upload	+3
Flooding activity	+3

Threat Levels

Score Range	Threat Level
0-1	Normal
2-3	Suspicious
4-5	Medium Threat
6+	High Threat

This intelligent scoring allows for nuanced threat assessment and automated actions beyond simple binary detection.



Why Behavioral Analysis?

Traditional IDS systems, relying solely on static signatures, often miss novel or polymorphic threats. Behavioral analysis bridges this gap understanding patterns and deviations.



Request Repetition

Detects unusual frequency or patterns of network requests, indicating brute-force attacks, DDoS attempts, or reconnaissance scans that might bypass signature checks.



Upload Attempts

Monitors abnormal upload volumes, unusual file types, or unauthorized destinations for outbound data, signaling potential data exfiltration or malware distribution.



Traffic Behavior

Identifies deviations from established network baselines, such as unexpected port usage, sudden spikes in traffic volume, or communication with suspicious external IP addresses.



Recurring Attack Patterns

Recognizes sequences of events that, individually might seem harmless, but collectively form a malicious campaign, providing early warning of sophisticated multi-stage attacks.

Automated Response: IPS Intervention

When threat thresholds are exceeded, our Intrusion Prevention System (IPS) intervenes immediately to neutralize threats without human intervention.



IP Extraction

Automatically identifies and extracts the source IP of the attacking entity.



Firewall Blocking

Integrates with iptables to block malicious IPs at the network perimeter.



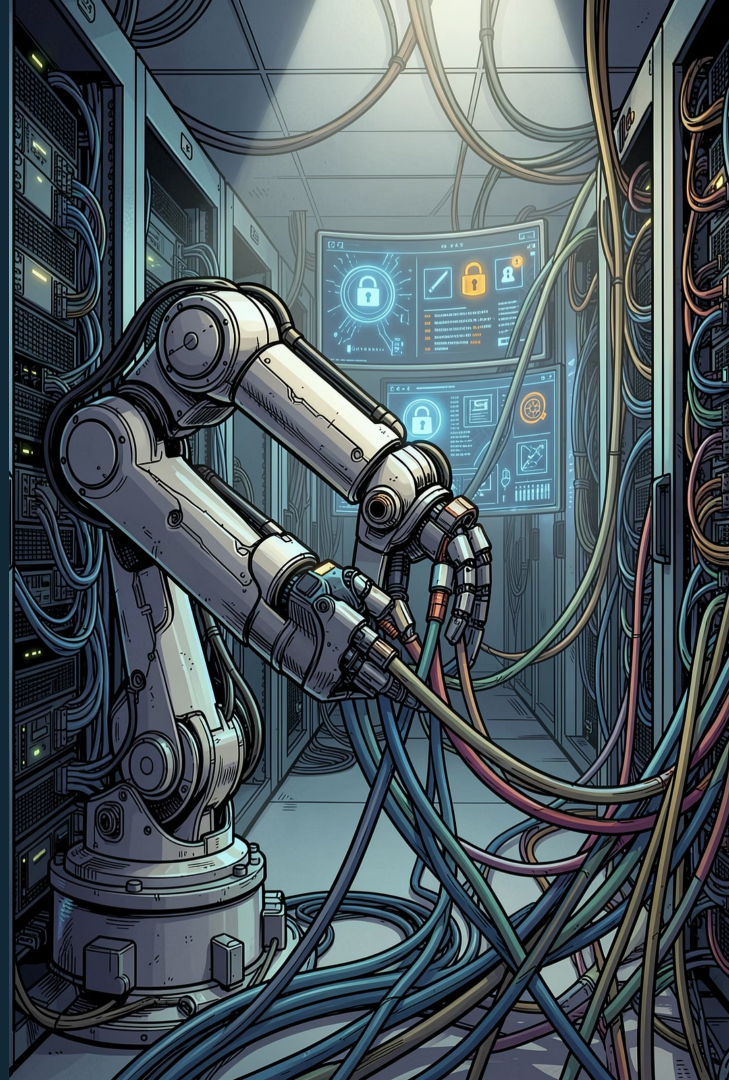
Generate Alerts

Sends real-time notifications to administrators about the detected threat and action taken.



Log Threats

Comprehensive logging of all threat details for forensic analysis and future pattern recognition.



Advanced Detection Mechanisms

Our platform leverages a multi-layered approach to detect a wide array of threats, combining established techniques with behavioral analysis.



Signature Detection

Identifies known threats by matching network traffic against a database of attack signatures and predefined rules.



ICMP Detection

Monitors ICMP traffic for anomalies like excessive requests, unusual payload sizes, or unsolicited replies, indicating reconnaissance or denial-of-service attempts.



HTTP Detection

Analyzes HTTP requests and responses for suspicious patterns, malformed headers, unusual user-agents, or attempts to exploit web vulnerabilities.



Suspicious File Uploads

Detects and flags uploads of potentially malicious file types (e.g., executables, scripts) or abnormally large files to sensitive server locations, preventing data exfiltration or malware injection.